

JEE MAIN REVISION MASTER: ALDEHYDE AND KETONE 2026 3 QUESTIONS 5% WEIGHTAGE

↑50.15%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Aldehyde And Ketone - Core Application	22 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Aldehyde And Ketone - Core Application (Very High Yield — 22 PYQs)

Core Principle: Key problem pattern in Kcet Aldehyde And Ketone.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The compound that does not answer iodoform test is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The major product 'A' in the given reaction is Benzaldehyde $\rightarrow$ $\leq$ farrow $\blacksquare$ 'A' (Major product)..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Match the reagents in List-I with products obtained from their carbonyl compounds in List-II. List-I List-II (a) NH_2OH (i..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The compound that does not answer iodoform test is

- (A) Ethanal
- (B) Acetone
- (C) Ethanoic acid
- (D) Acetophenone

Correct Answer: None

Solution Analysis: The compound that does not answer the iodoform test is: Option C: Ethanoic acid Iodoform test is given by compounds containing the group $-\text{COCH}_3$. This is called the methyl ketone group. Some alcohols which can be oxidised to methyl ketones also give this test. Now check the options: Ethanal has the structure: CH_3CHO . It gives iodoform test. Acetone has the structure: CH_3COCH_3 . It contains the methyl ketone group, so it gives iodoform test. Acetophenone has the structure: $\text{C}_6\text{H}_5\text{COCH}_3$. It also contains the methyl ketone group, so it gives iodoform test. Ethanoic acid